

M-DaQ: Retrieving Samples with Multilingual Diversity and Quality for Instruction Fine-Tuning Datasets

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1 MOTIVATION

- Multilingual IFT data is scarce, skewed toward English, and lacks systematic curation
- Llama-3 IFT dataset: only 3.01% multilingual samples
- No language-agnostic quality scoring method for multilingual data
- Data diversity studied only in English settings
- Superficial Alignment Hypothesis (SAH) unverified in multilingual contexts

Three Challenges Addressed:

- C1** No extensible quality scoring → QSM with triplet loss
- C2** Diversity selection English-only → DAS inspired by MMR
- C3** SAH unverified multilingually → 1K → 52K scale study, 8 langs

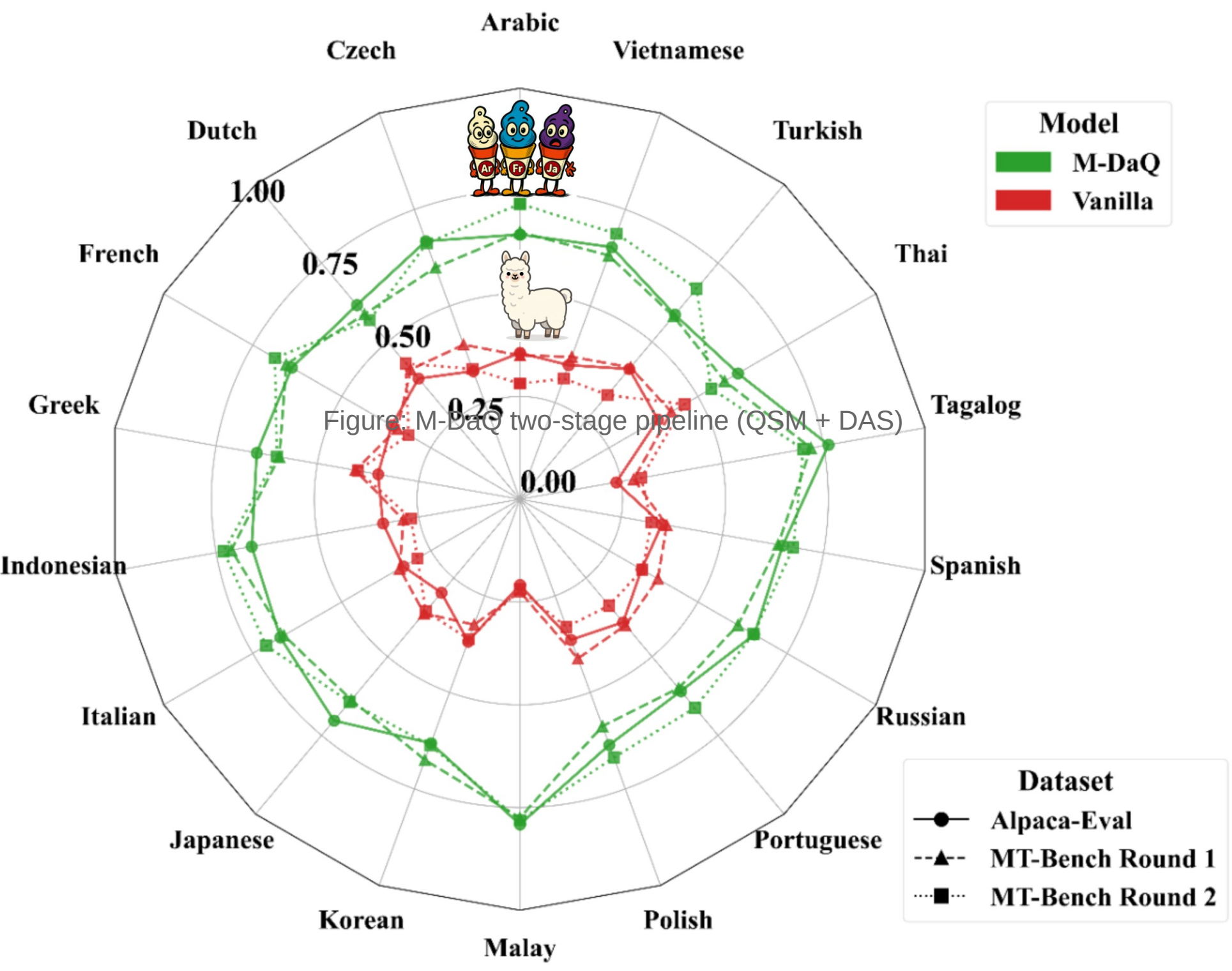
2 METHOD

Stage 1: Quality Scoring Model (QSM)

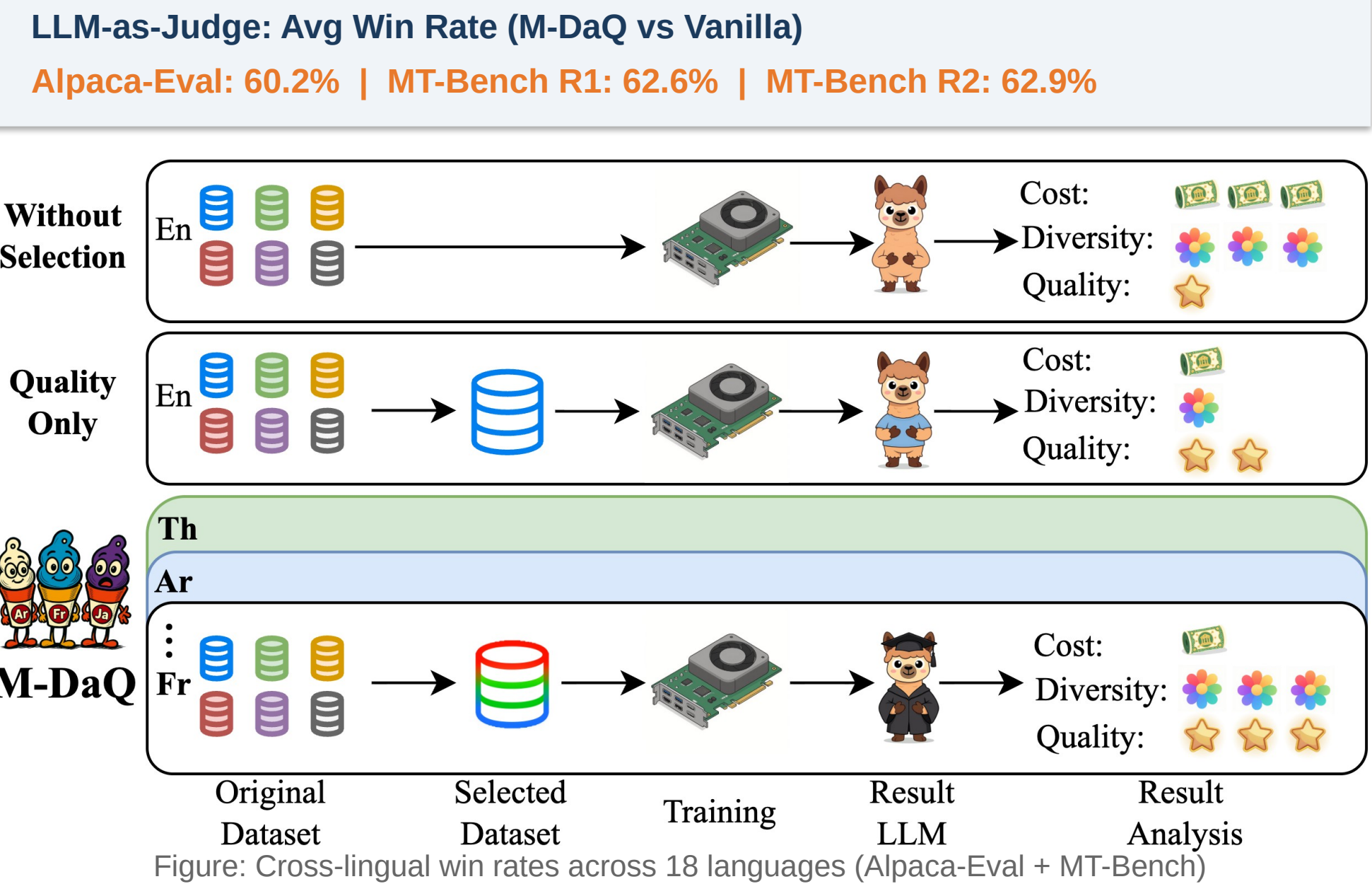
- Fine-tuned on ~2.3K expert-revised samples × 18 languages
- Triplet loss: instruction → positive (expert) vs negative (original/MT)
- Language-agnostic quality signal from embedding space

Stage 2: Diversity-Aware Selection (DAS)

- MMR-inspired: Quality ≈ Relevance, Diversity ≈ Novelty
- Two-stage pipeline: $O(n^2)$ → $O(n \log n)$
- Stage 1: Select top-n by QSM score (quality)
- Stage 2: Greedily augment from uncovered clusters (diversity)
- Ratio n_quality : n_diversity = 6:1 (empirically tuned)



3 KEY RESULTS



Key Observations:

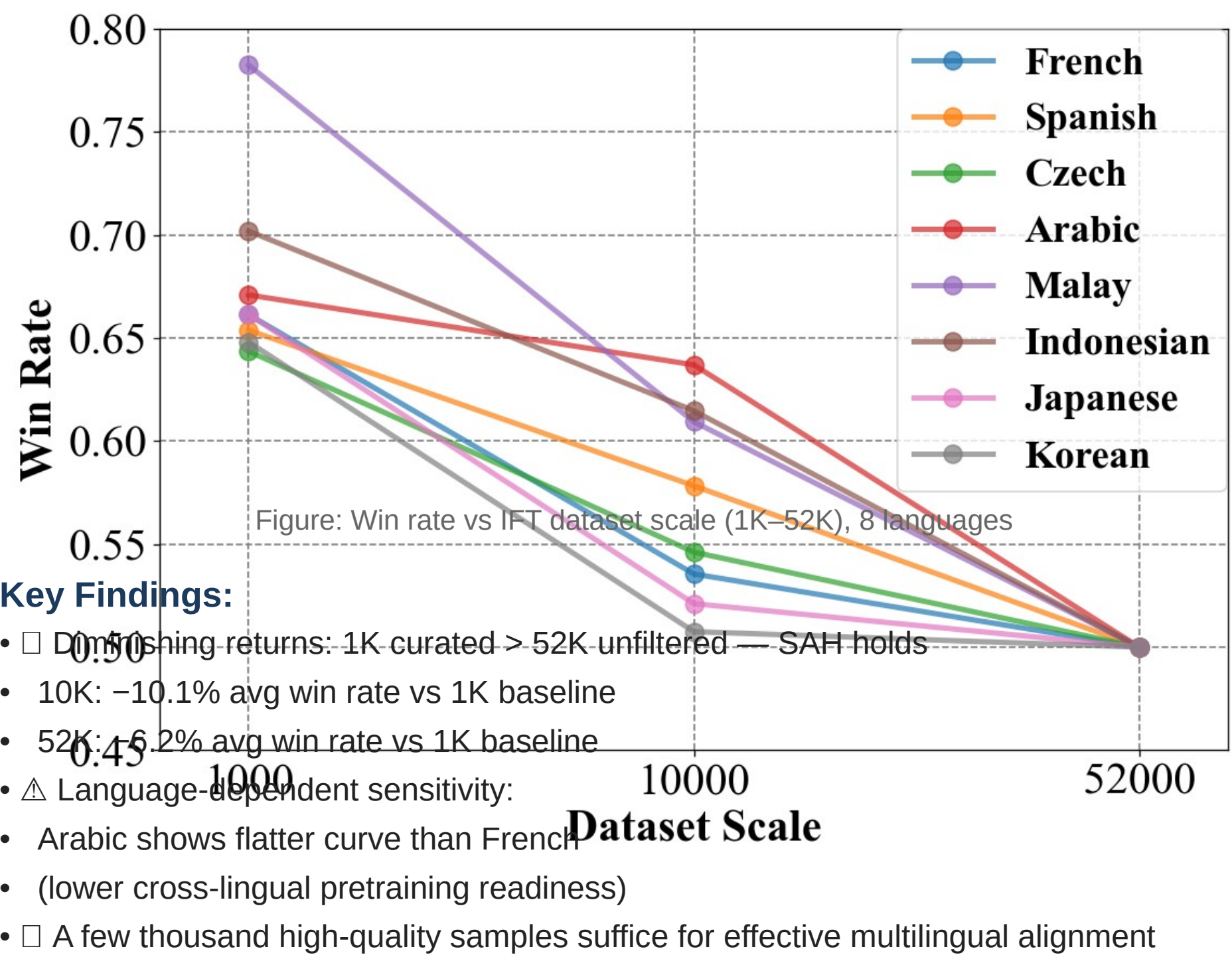
- Consistent improvement across all 18 languages
- Gains MORE pronounced in low-resource languages (Tagalog, Malay)
- M-DaQ effectively mitigates linguistic imbalance and data scarcity

Human Evaluation (6 Langs, 900 Samples, 58 Person-Hours):

Language	Alpaca-Eval	MT-Bench R1	MT-Bench R2
Japanese	86%	84%	80%
Korean	94%	94%	94%
Russian	70%	86%	84%
Portuguese	80%	78%	84%
Greek	58%	76%	82%
French	80%	92%	88%
Average	78.0%	85.0%	85.3%

4 SAH IN MULTILINGUAL SETTINGS

First systematic investigation of the Superficial Alignment Hypothesis across languages.



5 CULTURAL LOCALIZATION

Instruction:

Rédigez un blog de voyage captivant sur un voyage récent en Corse, en mettant en évidence les expériences culturelles et les attractions incontournables. (Write a captivating travel blog about a recent trip to Corsica, highlighting cultural experiences and must-see attractions.)

Response:

Faire un voyage en Corse est l'une des aventures ... (Making a trip to Corsica is one of the most fascinating and amazing adventures...)

L'une des expériences les plus mémorables de mon voyage a été mon visit à Piana (a historic mountain village in southern Corsica), un petit village perdu dans les montagnes de l'intérieur de l'île. Le village est fierement animé par sa population de langue corse (Corsican language) et sa cuisine traditionnelle (Corsican traditional cuisine). Conclusion of pastoral life, Mediterranean influences). Je'ai eu la chance de goûter certains des plats locaux, y compris la ravioli et la figatelli (a traditional Corsican dried pork sausage), et de les apprécier avec des vins rouges corsiques (Corsican wines).

Example: French travel query about Corsica → M-DaQ references Piana, figatelli, local conventions

- Human eval confirms gains in cultural relevance, contextual appropriateness, instruction-following
- First empirical validation of SAH in multilingual settings
- Code: github.com/zhaocorey/M-DaQ

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- Paper: [arXiv:2509.15549](https://arxiv.org/abs/2509.15549)
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